

Input & Encoding

Age	Education	Income
40	Year 12	>50K
28	?	>50K
?	Graduate	?
...	...	...

One-hot Encoding
*integer index embeddings

Edu1	Edu2	Edu3	...	Income
1	0	0	...	1
NaN	NaN	NaN	...	0
0	1	0	...	NaN
...	...	...	...	...

Age
40
28
?
...

x^{mis} in Continuous Variables

x^{mis} in Discrete Variables

Two-Channel Diffusion Imputation

Channel 1

Continuous Diffusion
Deterministic Sampling

Share x^{obs} + update $\hat{x}^{\text{obs}}$
Trained with
Self-masking Strategy

Conditional Latent-Path
Discrete Diffusion

Channel 2

Joint Reconstruction

Age	Education	Income
40	Year 12	Yes
28	Year9	No
35	Graduate	Yes
...	...	...

Reconstruct
Continuous
Variables

Reconstruct
Discrete Variables

Education	Income
Year 12	Yes
Year 12	No
Graduate	Yes
...	...

Decode

Via
ArgMax